

Research Interests

My research focuses on developing **human-like and trustworthy Artificial Intelligence** through the integration of computer vision, multimodal modeling, and data-driven problem solving. I am particularly interested in building **robust and interpretable vision-language systems** that can operate reliably in real-world environments, with emphasis on:

- (1) **Efficient and reliable visual understanding**, including object detection and lightweight architectures for real-time inference;
- (2) **Multimodal fusion** of visual, linguistic, and contextual signals for improved scene interpretation and decision-making;
- (3) **Behavioral and environmental analysis** from static and dynamic visual data to support real-world applications such as disaster assessment;
- (4) **Trustworthy AI**, ensuring fairness, robustness, and safe deployability of machine learning and large vision-language models.

Education

Sogang University (Transferred 2024)

Seoul, South Korea

B.S. IN COMPUTER SCIENCE

Expected Aug. 2026

- Relevant Coursework: Deep Learning, Machine Learning, Big Data Computing

Hankuk University of Foreign Studies (HUFS)

Seoul, South Korea

INFORMATION & COMMUNICATION ENGINEERING

2020 – 2023

- Transferred to Sogang University in 2024

Projects

MULTI-TRACK VIDEO-TO-AUDIO DESCRIPTION (4 MEMBERS)

Mar 2026 – Present

- Developing a structured multi-track audio description pipeline from video using Large Vision-Language Models (e.g., Gemini-Pro, GPT-4o), focusing on dialogue, sound effects, music, and ambience tracks. Proposing a reference-based evaluation and iterative refinement framework to improve temporal alignment and description consistency.

CHEST X-RAY MULTI-DISEASE DETECTION SYSTEM (CXr-CAD) (5 MEMBERS, TEAM LEADER)

Mar 2026 – Present

- Led a 5-member team and developed the core deep learning model (DenseNet-121) for a multi-label thoracic disease CAD system, addressing class imbalance with Focal Loss and enhancing interpretability via Grad-CAM.

RAG-BASED, AI-POWERED CRM MANAGER (5 MEMBERS, TEAM LEADER)

Sep 2025 – Dec 2025

- Architected a **Multimodal RAG** pipeline on GCP, integrating GPT-4o (Vision/Whisper) for unified processing of text and image customer data with ChromaDB vector storage.
- Engineered a **self-evolving loop** using PostgreSQL logs and dynamic prompting, enabling the agent to learn from interactions and align with owner policies.
- Deployed **autonomous agent tools** (e.g., CS replies, BI reports) through a full-stack architecture, featuring a FastAPI backend and an interactive frontend dashboard for admins and agents.
GitHub: github.com/suhwantcha/CS-Agent

EFFICIENTNET DOG & CAT CLASSIFICATION (4 MEMBERS)

Dec 2024

- Implemented image classification on the Oxford-IIIT Pet Dataset using transfer learning with **EfficientNet-B0**, optimizing data augmentation and regularization (CutMix) and achieving **88.08% validation accuracy**.
GitHub: github.com/suhwantcha/efficientnet-Image-classification

YOLO OBJECT DETECTION (YOLOv5, YOLOv7, AND YOLOv11)

Mar 2025

- Trained and evaluated YOLO models (**YOLOv5, YOLOv7, and YOLOv11**) for object detection on the PASCAL VOC dataset, including data pre-processing (XML to .txt), transfer learning, and performance evaluation using mAP, optimized for real-time detection.
GitHub: github.com/suhwantcha/yolo-object-detection

YOLO-PSYCHO-ANALYSIS (PSYCHOLOGICAL ANALYSIS FROM HOUSE SKETCHES)

Apr 2025 – May 2025

- Developed a **YOLOv8**-based model to analyze psychological characteristics by detecting elements such as doors, windows, sun, and chimney in user-drawn house sketches, converting them into quantitative scores and generating customized analysis scripts based on HTP test theory and art therapy.
GitHub: github.com/suhwantcha/YOLO-Psycho-Analysis

PINTOS OS

Sep 2025 – Dec 2025

- Implemented a **Pintos OS kernel** in C, including user program execution, file system support, and **priority-based thread scheduling**, with system calls, per-thread file descriptor management, and lock-based synchronization for safe concurrent file access.
GitHub: github.com/suhwantcha/OS-Pintos-Project

Experience

Teaching Assistant, Basic Artificial Intelligence Programming SOGANG UNIVERSITY Supported Python programming coursework by handling Q&A, grading assignments, and managing attendance.	Seoul, South Korea Mar 2026 – Jun 2026
Teaching Assistant, Understanding Artificial Intelligence SOGANG UNIVERSITY - Supported an Introduction to Artificial Intelligence course for 50+ students, managing attendance, Q&A, and grading. - Conducted PyTorch-based labs on deep learning architectures, including CNNs and Transformers.	Seoul, South Korea Sep 2025 – Dec 2025
Teaching Assistant, Computer Architecture SOGANG UNIVERSITY Supported course delivery for Computer Architecture and assisted 40+ students in understanding computer systems fundamentals.	Seoul, South Korea Mar 2025 – Jun 2025
Member & Mentor, Parrot Data Science Club SOGANG UNIVERSITY - Participated in data science and AI study sessions involving data analysis and deep learning projects. - Mentored 5 junior members (Mar 2025 – Jun 2025) in data analysis and CNN-based computer vision projects. - Organized internal Kaggle-style ML competitions and technical workshops.	Seoul, South Korea Sep 2024 – Present
Teaching Assistant, English & Mathematics MEGASTUDY RUSSEL & YESUM ACADEMY - Tutored 30+ students in analytical problem-solving and critical thinking. - Supported classroom management and lesson preparation for middle/high school students.	Seoul, South Korea Feb 2024 – Present
Satellite Operation Squad Leader REPUBLIC OF KOREA ARMY - Ensured stable communication across military branches by operating the Military Satellite Communication System. - Maintained mission stability and rapid response in combat readiness operations, ensuring uninterrupted communication services. - Led a team of 7 members, overseeing maintenance, inspection, and operation of satellite equipment to enhance mission readiness.	Daejeon, South Korea Jan 2022 – Jul 2023

Activities

Autonomous Taxi Design Project SOGANG UNIVERSITY - Designed an autonomous taxi system, focusing on AI-based navigation and safety protocols. - Developed cost-effective approaches for sensor fusion and environmental perception.	Seoul, South Korea Jul 2025 – Aug 2025
Runner-up, Sogang AI Data Analysis Runnerthon SOGANG UNIVERSITY - Built a real estate price forecasting model in a team of 4. - Achieved 2nd place among 15 teams using an XGBoost-based solution.	Seoul, South Korea Jan 2025 – Feb 2025
Software and AI Competency Enhancement Training Program REPUBLIC OF KOREA ARMY - Completed intensive training in software development and AI applications. - Developed a weather forecasting model using real-world data and machine learning techniques.	South Korea Jan 2023 – May 2023
English Training Program (R-EIP) HANKUK UNIVERSITY OF FOREIGN STUDIES (HUFS) Completed an intensive academic English program focused on communication skills.	Seoul, South Korea Oct 2021 – Dec 2021

Technical Skills

ML/DL Frameworks, PyTorch, TensorFlow, Hugging Face, LangChain, scikit-learn
Languages, Python, C/C++, SQL, R
Tools, Git, Linux, VS Code, Google Colab, MySQL, Jupyter Notebook, CLion, Notion
Data/Vision, Pandas, Numpy, Seaborn, OpenCV, Pillow, Cutmix

Languages

Korean, Native
English, Fluent (TOEFL iBT 100, Feb 2026)
Japanese, Basic (Studying)